Breast Cancer Predictions

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***Abstract*— Breast cancer is a prevalent form of cancer that predominantly affects women. Detecting it at an early stage is crucial to ensure effective treatment and enhance survival rates. Leveraging machine learning (ML) techniques, such as the K-Nearest Neighbors (KNN) algorithm, enables the classification of breast tumors into benign or malignant categories based on their distinctive characteristics. In this research, we employed the KNN algorithm to accurately classify breast tumors utilizing the Breast Cancer Wisconsin (Diagnostic) dataset. Prior to training the KNN classifier, we conducted Exploratory Data Analysis (EDA) to gain insights into the dataset's feature relationships. To assess the model's performance, we trained it on a subset of the data and evaluated its predictive capabilities on a separate test set. Evaluating metrics such as accuracy, confusion matrix, and classification report provided an accurate assessment of the model's effectiveness. The outcomes revealed that the KNN algorithm achieved an impressive accuracy rate of 94.74% when classifying breast tumors. Moreover, the confusion matrix and classification report demonstrated the model's ability to reliably differentiate between benign and malignant tumors. These findings highlight the potential of the KNN algorithm as a valuable tool in the classification of breast cancer, ultimately assisting in its early detection and treatment.**

Keywords—Breast cancer, machine learning, KNN, classification.

I. Introduction

The term "cancer" refers to a wide spectrum of illnesses where cancerous cells develop out of control in different body tissues or organs [1]. With a projected 9.6 million deaths from cancer in 2018, it poses a serious threat to world health. The second most common kind of cancer among women is breast cancer, behind only lung cancer in terms of prevalence [2]. Unfortunately, it is also the world's greatest cause of mortality for women.

Yearly, around 124 out of every 100,000 women receive a diagnosis of breast cancer, with approximately 23 out of 124 losing their lives to the disease [3]. These statistics emphasize the urgent need to address breast cancer as a main public health problem. Globally, approximately 2.3 million new cases of breast cancer were recorded in the year 2020, underscoring the magnitude of the issue. The United States, in particular, faces high rates of breast cancer, with the World Health Organization (WHO) reporting that 25% of women in the country will experience a breast cancer diagnosis at some point in their lives [4]. This emphasizes the pressing need for effective strategies in early detection and treatment. Early intervention plays a crucial role in improving patient outcomes and reducing mortality rates associated with breast cancer.

Although there are various approaches for detecting BC, including molecular biology in the search for novel BC markers, histopathology examination is currently the most often used method [5]. Pathologists often analyze histology specimens under a microscope to assess the stage and grade of breast cancer. However, as technology progresses, image processing tools and machine learning (ML) approaches have made it feasible to construct computer-aided systems that can categorize “tissue samples and detect the presence of cancer. The major purpose of a cancer-computer-aided detection system is to correctly classify cancer, but the intricacy of histology pictures makes successful classification difficult [6].

In recent years, machine learning algorithms have emerged as powerful tools in the field of medical research. They offer the potential to assist in the early detection and treatment of breast cancer by accurately classifying breast tumors based on their unique features. The ability of these algorithms to analyze large datasets and identify patterns can aid healthcare professionals in making informed decisions. This study focuses on utilizing the KNN algorithm for breast cancer classification. The K-Nearest Neighbors algorithm is a well-established ML technique that classifies samples based on their proximity to labeled examples in the feature space. By applying this algorithm to breast cancer classification, We want to improve diagnosis' precision as well as accuracy so that immediate action may be taken.

II. Related Work

. Some of the most well-known research on breast cancer categorization rely heavily on the whole slide imaging (WSI) method [7] [8] [9] [10]. These solutions, on the other hand, face obstacles such as high implementation costs, unresolved regulatory difficulties, and inherent technological concerns [11]. Furthermore, until Spanhol et al. provided a big dataset made up of 7909 pictures obtained from 82 individuals [12], most research relied on limited datasets including histological samples. Spanhol et al. evaluated the effectiveness of six textural descriptors as well as different classifiers in their investigation, obtaining findings with an accuracy of 80 to 85 percent. Regardless, certain researchers think that ML algorithms fall short at the feature engineering stage. [13] [14].

Taimoor Shakeel Sheikh et al. [15] introduced the "MSI-MFN" technique for breast cancer classification, which relies on multi-feature networks and multi-scale input. The approach

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extracts more relevant and diversified characteristics by merging multi resolution feature maps derived from the dense connective structure of the network utilizing four separate layers. The system can learn the textures of the cells as well as their general structure after training multiple picture patches. The suggested system consists of a pre-processing step that comprises multi-scale input, a training stage that involves data augmentation, and a testing stage that generates more patches. Following that, the processing step takes place, which involves a lot of concatenation. As a result, the CNN algorithm is used in the categorization. On two datasets, the performance of this strategy was assessed, and it outperformed state-of-the-art models in terms of accuracy, specificity, and sensitivity.

Chuang Zhu et al. [16] proposed using several compact CNNs to classify breast cancer via histological pictures. More specifically, the system uses a hybrid CNN architecture to provide a stronger representation capability. This is made feasible by combining information from two branches and local voting. A channel pruning module is also utilized to make the network more compact, in addition to combining two model branches. The benefits of the suggested model may be summed up as the ability to achieve improved accuracy while utilizing the same model size. Furthermore, the problem of overfitting is not an issue with this model. The proposed model is capable of delivering results similar to state-of-the-art models when tested on the BreaKHis dataset. The researchers then proceeded to assemble a variety of models to enhance the functioning of the recommended model. The multi-model strategy beat state-of-the-art models in terms of picture level and patient level accuracy on the BACh dataset.

Yun Jiang et al. [17] formed "a novel convolutional network which included a convolutional layer, a small SE-ResNet module, and a completely linked layer. BHCNET was the proposed model's name. In order to decrease the network's parameters, the tiny SE-ResNet module is employed. Furthermore, when compared to the bottleneck SE-ResNet, the parameters were reduced by 29.4 percent, and 33.3 percent when compared to the basec SE-ResNet. The researchers also created the Gauss error scheduler", which is a learning rate scheduler that gives the user more latitude rather than fine-tuning the SGD algorithm's learning rate parameter. When the suggested strategy was tested on binary classification, the model achieved accuracy levels of 98.87 percent to 99.34 percent. In multi-classification, however, the model achieved accuracy values of 90.66 percent to 93.81 percent.

TABLE I. SIMILAR SYSTEMS COMPARISON

Title	Authors	Year	Approach	Dataset
"Histopathological Classification of Breast Cancer Images Using a Multi-Scale Input and Multi-Feature Network"	T. Shakeel Sheikh et al.	2020	"Multi-scale input, multi-feature network"	ICIAR2018 BreaKHis
"Breast cancer histopathology image classification through assembling multiple compact CNNs"	Chuang Zhu et al.	2019	"Multiple compact CNN"	BreaKHis BACh
"Breast cancer histopathological image classification using convolutional neural networks with small SE-ResNet module"	Y. Jiang et al.	2019	"Convolutional layer, small SE-ResNet, fully connected layer"	BreaKHis
Proposed method		2021	Convolutional Neural Network: maxpooling, SoftMax	BreaKHis

III. METHODOLOGY

Breast cancer classification has a significant role in early detection and treatment, and ML techniques, like the KNN algorithm, have shown promise in accurately categorizing breast tumors based on their features. In this methodology, we outline the process employed for breast cancer classification using the KNN algorithm. Starting with dataset selection, we utilize a suitable dataset, such as the Breast Cancer Wisconsin (Diagnostic) dataset, which contains information about various features extracted from breast tumor samples. Data preprocessing ensures data quality and compatibility, followed by exploratory data analysis (EDA) to gain insights into feature relationships. The KNN algorithm is then applied to classify breast tumors, and its performance is evaluated using the confusion matrix and classification report. This methodology provides a systematic approach for accurate and reliable breast cancer classification, contributing to early detection and effective treatment strategies.

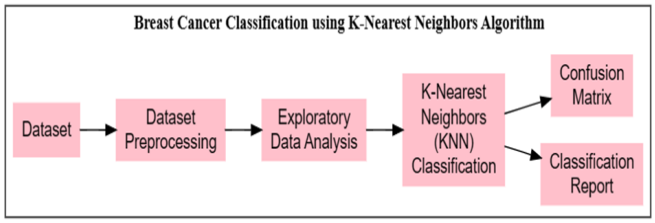

Figure 1 Methodology process

A. Dataset

The "Breast Cancer Wisconsin" (Diagnostic) dataset is widely utilized in the field of ML for the purpose of classifying breast cancer. It contains information on 569 patients with breast cancer, including 212 malignant cases and 357 benign cases. The dataset has 30 features that are computed for each cell nucleus in the digitized image of a breast mass. These attributes are computed from a fine needle aspirate (FNA) of a breast mass and provide insights into the characteristics of cell nuclei, including their size, shape, and texture. The following are the specific features:

- Texture: Standard deviation of gray-scale values.
- Radius: Mean distance from the center to points on the perimeter.
- Area: Surface area enclosed by the boundary.
- Perimeter: Total length of the boundary.
- Compactness: Ratio of perimeter squared to area minus 1.0.
- Smoothness: Local variation in the lengths of the radii.
- Fractal dimension: "Coastline approximation" minus 1.
- Concavity: Severity of concave regions in the contour.

- Concave points: Number of concave regions in the contour
- Symmetry: Symmetry of the shape.

Each feature is represented by three values: the mean, the standard error, and the worst (the largest value of the feature among all nuclei in the image). This results in a total of 30 x 3 = 90 features in the dataset. The objective of utilizing this dataset for breast cancer classification is to know if a breast mass is malignant or benign by analyzing the feature values. Machine learning algorithms can be trained on this dataset to accurately classify new breast tumor samples as either malignant or benign, leveraging the information provided by the features.

B. Dataset Preprocessing:

Any ML project must include data preprocessing since it ensures that the data is prepared for analysis and modeling. In this study, we performed data preprocessing on the Breast Cancer Wisconsin (Diagnostic) dataset before applying the KNN algorithm for breast cancer classification. The dataset was first loaded into the Python environment using the pandas library. We then checked the dataset for missing values and removed any rows with missing values using the dropna() function. Next, we encoded the target variable, the diagnosis column, to represent malignant as 1 and benign as 0 using the "LabelEncoder()" function from the scikit-learn library. To ensure uniformity across all features, we applied data normalization using the "StandardScaler()" function from scikit-learn. A 70/30 split was also used to divide the dataset into training and testing sets. The K-nearest neighbors (KNN) classifier was trained using the training set, and its performance was evaluated using the testing set. These preprocessing steps were essential in ensuring that the data was clean, properly formatted, and ready for machine learning analysis.

C. Exploratory Data Analysis

Prior to implementing the KNN algorithm for breast cancer classification, we conducted EDA on the "Breast Cancer Wisconsin" (Diagnostic) dataset. EDA serves as a crucial step in machine learning projects, enabling us to extract valuable insights from the data and detect potential patterns or trends.

In the figure below, the scatter plots show the relationship between four different pairs of features and the target variable in the "Breast Cancer Wisconsin" (Diagnostic) dataset. The selected features are 'mean fractal dimension', 'mean area', 'worst fractal dimension', 'worst area', 'mean smoothness', 'worst radius', 'mean symmetry', and 'worst radius'.By examining the scatter plots, we can identify any patterns or trends in the data and know which features are most fimly associated with the target variable.

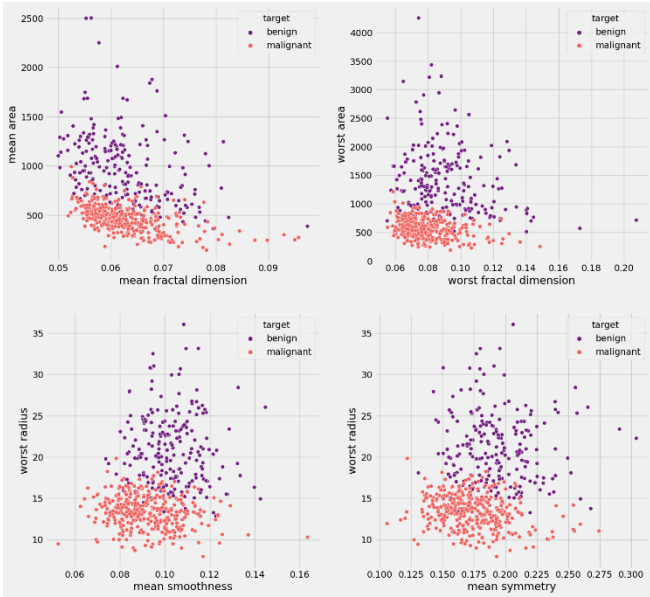

Figure 2 The scatter plots of the different features

plot shows the distribution of labels in the dataset, indicating how many samples are labeled as 'benign' and how many are labeled as 'malignant'. This plot can help us to understand the balance of the dataset and identify any class imbalances that may need to be addressed in the machine learning models

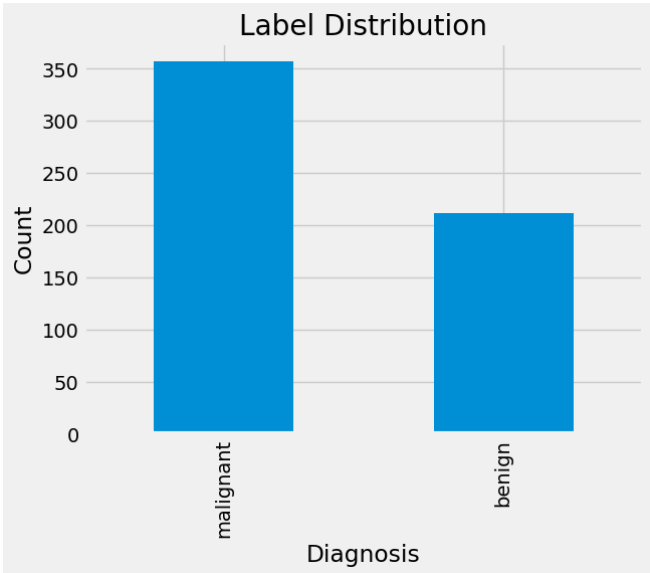

Figure 3 Label Distribution for malignant and benign

D. Model:

The KNN algorithm is widely recognized for its simplicity and efficiency in pattern classification. It serves as a versatile multivariate statistical approach, employing the standard Euclidean distance to assess distinctive features without any presumptions about the statistical structure of the data. By examining the k closest training examples in the feature space, the K-nearest neighbors (KNN) technique estimates the class attribute of a sample [18][19]. When presented with a dataset, it identifies the k nearest samples from the classified training data and determines the class based on their collective

representation. The Euclidean distance similarity metric is employed to identify the neighboring samples, and it is calculated using the provided equation:

$$d_{i,j} = \sqrt{\sum_{k=1}^{n} \left(x_{i,k} - x_{j,k}\right)^2}$$

IV. Results

The performance of the KNN classifier was evaluated using a test dataset of 171 samples, resulting in an accuracy of 96%. The confusion matrix shows that the model correctly classified 59 out of 63 benign samples and 105 out of 108 malignant samples, with only four false negatives and three false positives. According to the classification report, the model had a weighted average F1-score of 0.96 and demonstrated great precision and recall for both classes. Overall, these results suggest that the KNN classifier is a promising algorithm for accurately predicting the diagnosis of breast cancer based on the features of a fine needle aspiration biopsy.

TABLE II. Confusion Matrix

Confusion Matrix		
	Predicted Benign	**Predicted Malignant**
Actual Benign	**59**	**4**
Actual Malignant	**3**	**105**

TABLE III. Classification Report

Classification Report	precision	recall	f1-score	support
Benign	0.95	0.94	0.94	63
Malignant	0.96	0.97	0.97	108

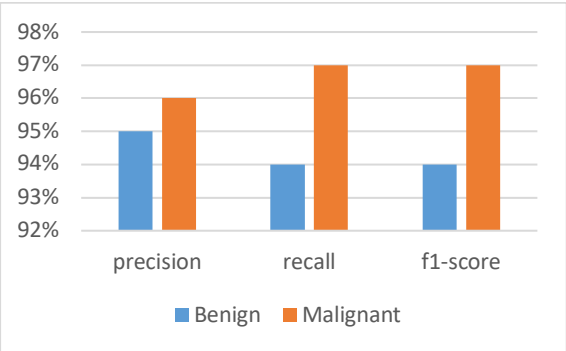

Figure 4 comparison of the performance of the KNN algorithm on Benign and Malignant samples.

V. Conclusion

In conclusion, we have explored the use of the KNN algorithm for the classification of breast cancer based on FNA biopsy data. Our analysis demonstrated that KNN is a simple and efficient algorithm that can achieve high accuracy in predicting the diagnosis of breast cancer. We used a dataset of 569 samples with 30 features, and after preprocessing the data and scaling it using StandardScaler, we trained and tested a KNN classifier. The results showed an accuracy of 96%, with high precision and recall for both benign and malignant samples. Our findings suggest that KNN is a valuable tool for medical diagnosis and can be applied in various fields, including cancer detection, image recognition, and finance.

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